Skills Summary

Transformation Evidence for Congruence and Similarity

Use this reference while completing your worksheet or when studying.

Skill 1 — Congruence evidence: equal lengths

Distance between two points:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Translations, reflections and rotations are **rigid motions**: they never change a length. So if a figure maps onto another by rigid motions, every pair of matching sides is equal and the figures are congruent.

Example: Find the length of the segment from $(1, 2)$ to $(4, 6)$.
Step 1. Two coordinate points and a length wanted — that is the distance formula.
Step 2. Run $= 4 - 1 = 3$, rise $= 6 - 2 = 4$, so $d = \sqrt{9 + 16} = \sqrt{25}$. $\Rightarrow$ $d = \mathbf{5}$

Try it: Find the length of the segment from $(0, 1)$ to $(6, 9)$.

check: **10**

Skill 2 — Similarity evidence: one constant ratio

Rule: $k = \dfrac{\text{image length}}{\text{matching original length}}$

Figures are **similar** when every pair of matching sides gives the *same* k. They are **congruent** only in the special case $k = 1$. One pair giving a different ratio ends the argument: the figures are neither.

Example: A side of 6 matches a side of 9. Find the ratio.
Step 1. Similarity is judged by a ratio, so divide the image length by the one it matches.
Step 2. $k = \dfrac{9}{6}$, and dividing top and bottom by 3 gives $\frac{3}{2}$. $\Rightarrow$ $k = \boldsymbol{\frac{3}{2}}$

Try it: A side of 10 matches a side of 4. Find the ratio.

check: $\boldsymbol{\frac{2}{5}}$

Skill 3 — Slope evidence: sides stay parallel

Slope: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$

A dilation multiplies every rise and every run by the same k, so slopes are unchanged and each image side is parallel to the side it came from. Parallel matching sides mean equal matching angles — the angle half of the similarity argument.

Example: Find the slope of the segment from $(1, 2)$ to $(5, 4)$.
Step 1. We want the direction of the segment, not its length, so use slope.
Step 2. $m = \dfrac{4-2}{5-1} = \dfrac{2}{4}$, which reduces. $\Rightarrow$ $m = \mathbf{\frac{1}{2}}$

Try it: Find the slope of the segment from $(2, 5)$ to $(6, 2)$.

check: $-\mathbf{\frac{3}{4}}$

Skill 4 — Naming the transformation

Coordinate rules:

reflection in the x-axis: $(x, y) \mapsto (x, -y)$ • reflection in the y-axis: $(x, y) \mapsto (-x, y)$
rotation of 180° about the origin: $(x, y) \mapsto (-x, -y)$ • translation: add the same pair to every point

Example: Reflect $(3, 5)$ in the x-axis.
Step 1. Compare the wanted map with the rules above: reflecting in the x-axis keeps x and flips y.
Step 2. The x-coordinate stays 3; the y-coordinate becomes -5. $\Rightarrow$ $\mathbf{(3, -5)}$

Try it: Rotate $(-2, 4)$ by 180° about the origin.

check: $\mathbf{(2, -4)}$

(!) Watch out: equal ratios prove *similar*, not *congruent*. Congruent is the special case where the ratio is 1 and the lengths themselves match.